

DSA0502 – Query Processing for Data Science	
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LAB PROGRAMS 23 AND 24

23. Draw a Line Using Axis Values Taken from a Text File

Question

Write a Python program to draw a line using given axis values taken from a text file, with suitable labels on the x-axis, y-axis, and a title.

Aim

To read axis values from a text file and plot them as a line graph.

Dataset — data.txt

```
%%writefile data.txt
```

```
1 5
```

```
2 10
```

```
3 15
```

```
4 20
```

```
5 25
```

Python Program

```
import matplotlib.pyplot as plt
```

```
# Create empty lists
```

```
x = []
```

```
y = []
```

```
# Read data from text file
```

```
with open("data.txt", "r") as file:
```

```
    for line in file:
```

```
        a, b = line.split()
```

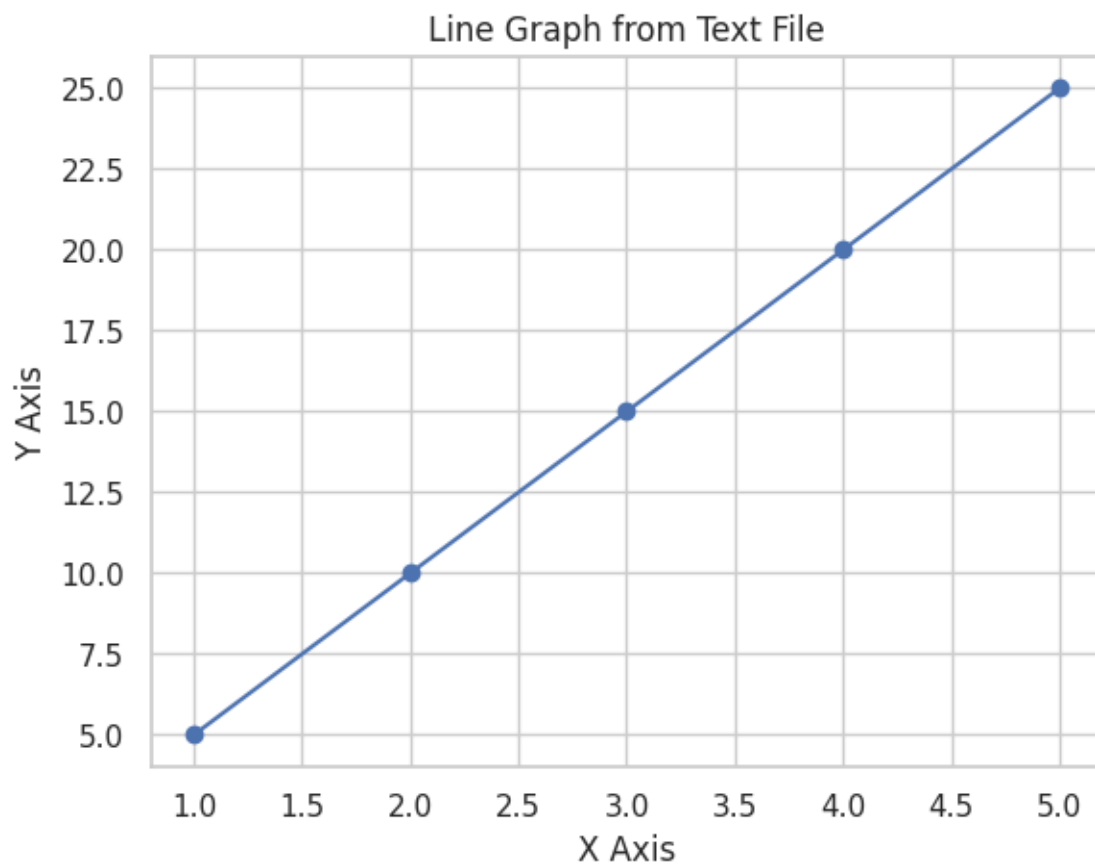
```
        x.append(int(a))
```

```
        y.append(int(b))
```

```
# Plot the line
plt.plot(x, y, marker="o")
# Add labels and title
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Line Graph from Text File")
```

```
# Display the graph
plt.show()
```

Output



Result

The data from the text file was successfully read and plotted as a line graph.

24. Draw Line Charts of Financial Data

Question

Write a Python program to draw line charts of the financial data.

Aim

To visualize financial data using line charts.

Dataset — financial_data.csv

```
%%writefile financial_data.csv
```

```
Month,Revenue,Profit
```

```
Jan,5000,1200
```

```
Feb,6500,1800
```

```
Mar,7000,2200
```

```
Apr,8000,2600
```

```
May,9000,3200
```

Python Program

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read the dataset
```

```
df = pd.read_csv("financial_data.csv")
```

```
print("Financial Dataset:")
```

```
print(df)
```

```
# Plot Revenue and Profit
```

```
plt.plot(df["Month"], df["Revenue"], marker="o", label="Revenue")
```

```
plt.plot(df["Month"], df["Profit"], marker="o", label="Profit")
```

```
# Add labels and title
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Amount")
```

```
plt.title("Financial Data")
```

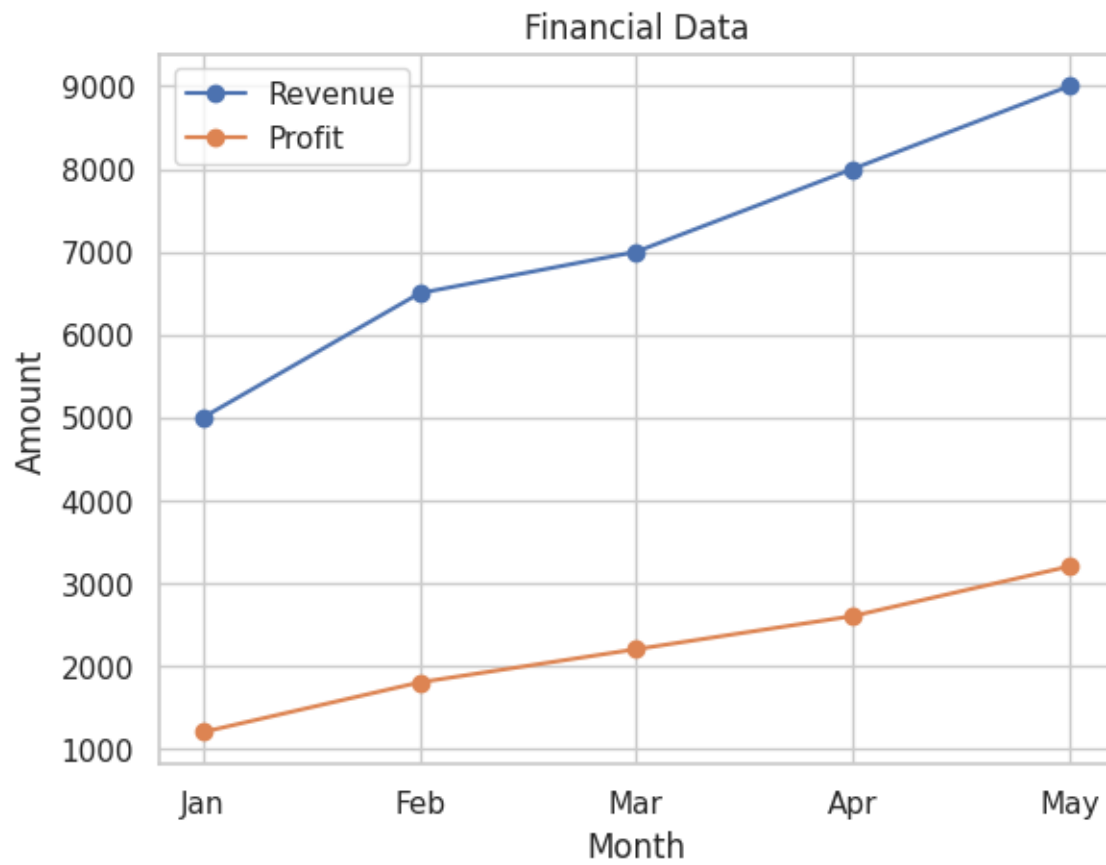
```
# Display legend and graph
```

```
plt.legend()
```

```
plt.show()
```

Output

```
... Financial Dataset:
   Month  Revenue  Profit
0    Jan     5000     1200
1    Feb     6500     1800
2    Mar     7000     2200
3    Apr     8000     2600
4    May     9000     3200
```



Result

The financial data was successfully visualized using line charts.